

ZOUBIRI Wissal Zeyneb

To my family – my first and most enduring support.

To Wissal – for her patience, her creativity, and her friendship.

BENATTIA Djouhina

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****List of Acronyms****

Acronym	Definition	Acronym	Definition
----	----	----	----
ACK	Acknowledgment	**LRU**	Least Recently Used
AODV	Ad hoc On-demand Distance Vector	**MANET**	Mobile Ad hoc Network
API	Application Programming Interface	**NFR**	Non-Functional Requirement
BAN	Body Area Network	**OLSR**	Optimized Link State Routing Protocol
BLE	Bluetooth Low Energy	**P2P**	Peer-to-Peer
BTS	Base Transceiver Station	**RFC**	Request for Comments
DHCP	Dynamic Host Configuration Protocol	**RREP**	Route Reply (AODV)
DTN	Delay-Tolerant Network	**RREQ**	Route Request (AODV)

	ECDH		Elliptic Curve Diffie-Hellman			**SDK**		Software Development Kit	
	EOF		End-of-File delimiter			**SOS**		Save Our Souls (international emergency signal)	
	FR		Functional Requirement			**SPAN**		Smartphone Ad hoc Network	
	GO		Group Owner (Wi-Fi Direct)			**SQL**		Structured Query Language	
	GPS		Global Positioning System			**SRP**		Smart Routing Protocol	
	ID		Identifier			**TCP**		Transmission Control Protocol	
	IEEE		Institute of Electrical and Electronics Engineers			**TTL**		Time To Live	
	IP		Internet Protocol			**UI**		User Interface	
	IoT		Internet of Things			**UUID**		Universally Unique Identifier	
	JSON		JavaScript Object Notation			**WFD**		Wi-Fi Direct	

General Introduction

At 04:17 on February 6, 2023, a 7.8-magnitude earthquake struck Kahramanmaraş in southern Turkey. Within minutes, this pattern is not unique to Turkey. The 2010 Haiti earthquake, the 2003 Boumerdès earthquake in Algeria,

Context and Motivation

This thesis is developed within the Master's program in Computer Science at the University Mohamed Boudiaf

Problem Statement

The central research question is: how can a group of standard Android smartphones, without internet connection,

Objectives

- Design a four-layer communication architecture for infrastructure-free smartphone mesh networking;
- Implement a hybrid BLE+Wi-Fi Direct discovery and connection mechanism with automatic group formation;
- Design and implement the Smart Routing Protocol (SRP) with TTL enforcement, LRU deduplication, and prioritization;
- Build a multi-group bridge topology supporting up to 20 devices without root access or specialized hardware;
- Develop an Android application with a stress-tolerant tactical interface across five functional screens;
- Validate the system through real-device testing and document undocumented platform behaviors.

Thesis Structure

This document is organized into three chapters. Chapter 1 surveys the state of the art: documented disaster response

Chapter 1: State of the Art

1.1 Introduction

This chapter surveys the state of the art across three dimensions. First, the documented failure patterns

1.2 Emergency Communication: The Infrastructure Dependency Problem

1.2.1 Documented Failure Patterns

The foundational study by Quarantelli [1], drawing on six decades of disaster case studies, consistently

- Physical destruction of Base Transceiver Stations (BTS) by earthquake impact, floodwater, or blast overpressure;
- Power grid failure eliminating diesel generator backup at switching centers, typically within 4-8 hours;
- Network congestion: simultaneous emergency calls from millions of users overwhelming remaining switching capacity;
- Backbone fiber-optic cable severance cutting connectivity between the affected zone and the national network;
- Electromagnetic interference from emergency vehicle radios disrupting remaining wireless base stations.

The 2011 Great East Japan Earthquake – the most extensively documented case – disrupted 1.5 million fixed-line

1.2.2 Requirements for Infrastructure-Free Emergency Communication

Synthesizing the disaster response literature [1,2] and the operational specifications of search-and-rescue

ID	Requirement	Target	SPAN RESCUE
:---	:---	:---	:---
R1	Zero-infrastructure	No network, AP, or server	BLE \+ Wi-Fi Direct only
R2	Rapid deployment	< 10 seconds, no config	Auto-discovery \+ formation

R3	Multi-hop relay	Beyond direct radio range	SRP up to 10 hops
R4	SOS priority	Delivery before all other traffic	Priority queue P=1
R5	Battery efficiency	Minimal energy in background	BLE BALANCED mode
R6	Standard hardware	Consumer smartphones only	Android API 29+
R7	Self-healing	Recover from node failure	Heartbeat \+ re-election

Table 1.1: Emergency communication requirements and SPAN RESCUE's response to each

1.3 Mobile Ad hoc Networks: Theoretical Foundations

1.3.1 Definition and Core Characteristics

A Mobile Ad hoc Network (MANET) is formally defined in IETF RFC 2501 \[3\] as a self-configuring, infrastr

- Dynamic topology: nodes continuously join, leave, and move, making any network state snapshot obsolete w
- Distributed control: no authority maintains routing tables or assigns addresses; all coordination is eme
- Constrained resources: battery-powered nodes have limited CPU, RAM, and radio bandwidth – overhead must
- Multi-hop delivery: most node pairs are beyond direct radio range; messages require relay through intern

These characteristics define the core routing challenge: deliver messages reliably across a topology chang

1.3.2 Routing Protocol Families

Three routing protocol families address this challenge with different trade-offs between accuracy (requiri

****Proactive protocols**** maintain continuously updated routing tables through periodic topology broadcasts.

****Reactive protocols**** discover routes only when needed. AODV (RFC 3561 \[5\]) floods a Route Request (RRE

****Flooding-based approaches**** broadcast every message to all reachable nodes, guaranteeing delivery as lon

Protocol Family	Representative	Overhead	Key Trade-off
-----	-----	:---:	-----
Proactive (table-driven)	OLSR \[4\], DSDV \[18\]	High – periodic floods	Zero latency; stale in hig
Reactive (on-demand)	AODV \[5\], DSR \[19\]	Medium – on demand	Accurate when stable; RREQ latency;
Flooding \+ mitigation	Epidemic, SRP (this work)	Low – dedup+TTL	Delivery guarantee; robust to fai
Hybrid	ZRP, HARP	Low-Medium	Zone-based; complex implementation overhead

Table 1.2: MANET routing protocol families – overhead characteristics and trade-offs

1.3.3 The Broadcast Storm Problem

Ni et al. \[6\] formally analyzed the broadcast storm problem in their seminal 1999 MobiCom paper. In a de

- Counter-based (TTL): each node maintains a hop counter per message; after maxHops relays, the packet is
- ID-based deduplication: each node maintains a cache of recently seen message IDs; packets whose ID is al
- Probabilistic/timed forwarding: each node applies a random delay before retransmission; this desynchroni

SPAN RESCUE's SRP implements all three mechanisms, as detailed in Chapter 2\.

1.4 Smartphone Ad hoc Networks (SPAN)

1.4.1 The SPAN Paradigm and Android Constraints

The SPAN paradigm recognizes that consumer smartphones already contain all the hardware needed for ad hoc

- Star topology constraint: within a single Wi-Fi Direct group, all client-to-client traffic must pass thr
- Single-group constraint: an Android device cannot participate in two Wi-Fi Direct groups simultaneously

1.4.2 Wi-Fi Direct – Architecture and Capabilities

Wi-Fi Direct (IEEE 802.11p2p) \[9\], certified by the Wi-Fi Alliance since 2010 and available on Android s

1.4.3 Bluetooth Low Energy – Energy-Efficient Discovery

BLE, introduced in Bluetooth 4.0 (2010), broadcasts 31-byte advertising packets on three dedicated channel

1.5 Survey of Existing SPAN Applications

1.5.1 FireChat (Open Garden, 2014–2020)

FireChat was the first widely deployed commercial SPAN application. It reportedly achieved 400,000 downloads.

1.5.2 Bridgefy

Bridgefy achieved prominence during the 2019 Hong Kong protests and 2019 Myanmar elections. A rigorous security audit [11] identified several vulnerabilities.

1.5.3 Meshtastic

Meshtastic implements a proper mesh routing protocol on dedicated LoRa radio hardware (433 MHz / 915 MHz), but lacks many features.

1.5.4 SmartGroup@Net (SGN)

Developed at the Research and Academic Computer Network in Warsaw, SGN [7] is the closest academic predecessor to SPAN RESCUE.

Capability	SPAN RESCUE	FireChat	Bridgefy	Meshtastic	SGN [7]
No infrastructure	✓	✓	✓	✓	✓
Standard hardware only	✓	✓	✓	✗ (LoRa)	✓
Multi-hop relay	✓ (10h max)	Limited	Limited	✓	✗ (1 hop)
Multi-group bridge	>8 dev	✓ (20 dev)	✗	✗	✓
SOS priority queue	✓ P=1, 3×	✗	✗	Partial	Partial
Store-and-Forward	✓ `shared_preferences` JSON	✗	✗	✓	✗
Live topology view	✓	✗	✗	✓	✗
Hop-count display	✓ per msg	✗	✗	✓	✗
Network self-healing	✓ <20s	✗	✗	✓	✗
End-to-end security	Planned	Unknown	Vulns [11]	✓	Partial
Open / auditable	✓	✗	Partial	✓	Research
Undocumented API docs	✓ (4 findings)	✗	✗	N/A	✗

Table 1.3: Survey of existing SPAN applications – features and identified gaps

1.6 Identified Gaps and Our Contribution

The survey reveals six capabilities that no existing solution fully provides simultaneously. SPAN RESCUE addresses the following gaps:

- Genuine multi-group star topology (>8 devices) with automated software bridge management connecting up to 100 devices.
- SOS signal priority (P=1) with TTL-bounded guaranteed delivery repeated 3× across the entire multi-group.
- Real-time mesh topology visualization with color-coded node status, hop counts, and role badges;
- Persistent store-and-forward using `shared\_preferences` JSON queue storage for temporarily disconnected nodes.
- Documented, auditable implementation including four undocumented Android Wi-Fi Direct and BLE API behaviors.
- All of the above on standard consumer Android hardware (API 29+), requiring no root access, no specialized hardware, and no network access.

1.7 Conclusion

This chapter has established the concrete motivation for infrastructure-free emergency communication through mesh networks.

Chapter 2: System Design and Conception

2.1 Introduction

This chapter presents the complete design and formal conception of SPAN RESCUE. It covers: system requirements, system architecture, and implementation details.

2.2 System Requirements

2.2.1 Functional Requirements

ID	Name	Full Description
FR1	Auto BLE Discovery	Detect nearby devices using BLE advertising+scanning within 5s of launch, with timeout.
FR2	Auto Group Formation	Form a Wi-Fi Direct star group with automatic GO election via Intent Value.
FR3	Multi-hop Relay	Forward messages up to maxHops=10 relay hops via SRP controlled flooding across the mesh.
FR4	SOS Broadcast	Broadcast SOS message with GPS coordinates and battery level at priority P=1; bypass hop limit.
FR5	Store-and-Forward	Persist messages for unreachable nodes in `shared_preferences` JSON queue; delete on success.
FR6	Live Topology	Render real-time mesh graph with color-coded node status, hop count, IP, and role badges.
FR7	Background Operation	Continue BLE discovery, TCP connections, SRP routing, and heartbeat when app is closed.


```
|
|         TCPUtls.send(entry.getValue(), msg); |
|         Log.i(TAG, "Bridge fwd → " \+ adjacentGroupId |
|             \+ " hop=" \+ msg.hopCount); |
|     } else { |
|         Log.d(TAG, "Group " \+ adjacentGroupId \+ " already seen, skip"); |
|     } |
| } |
| } |
```

2.4.4 Complete 20-Device Topology

Group	GO Node	Client Nodes	Bridge Node(s)	Connects To
A	GO-A	A1, A2, A3, A-Bridge	A-Bridge	Group B (port 8889\)
B	GO-B	B1, B2, B-Br-A, B-Br-C	B-Br-A / B-Br-C	Groups A & C
C	GO-C	C1, C2, C-Br-B, C-Br-D	C-Br-B / C-Br-D	Groups B & D
D	GO-D	D1, D2, D3, D4	D4 (SOS demo node)	Group C only

Table 2.8: Multi-group bridge topology – 20-device group assignment

Worst-case path (A1 → D3): A1 → GO-A (h1) → A-Bridge (h2) → GO-B (h3) → B-Br-C (h4) → GO-C (h5) → C-B

2.4.5 Network Self-Healing Scenarios

Failure Scenario	Detection	Automatic Recovery
Client disconnects mid-session	Heartbeat timeout 15s on GO	GO removes from registry; BLE monitors re
GO device loses power	TCP connection drop (immediate)	Clients detect drop; restart BLE+WFD discovery
Bridge node fails	Heartbeat timeout on both adjacent GOs	Groups isolate to independent stars; each c
New device joins existing mesh	BLE advertisement detected by nearest node	Nearest GO invites via WFD
GO election tie (equal Intent)	WFD stack reports conflict	Android WFD retries with randomized Intent

Table 2.10: Network self-healing – failure scenarios and automatic recovery

2.5 SRP Message Structure

Field	Type / Constraint	Full Description
messageId	String (UUID v4); required	Primary key for LRU SeenCache deduplication; generated at source
senderId	String XXXX-YYYY; required	Source device ID generated at install; stored in SharedPreferences
senderName	String max 20 chars; required	User-chosen display name shown in chat and topology screens
recipientId	String; required	Target device ID, or literal "BROADCAST" for group-wide delivery to all
messageType	Enum; required	CHAT SOS LOCATION HEARTBEAT ACK GO_BEACON
payload	String; nullable	Message text (CHAT); GPS lat,lng (SOS/LOCATION); timestamp (HEARTBEAT); JSON (ACK)
hopCount	int ≥ 0; initialized 0	Incremented by 1 at each relay node BEFORE forwarding; never decreme
maxHops	int 1-20; default 10	TTL ceiling; message silently discarded when hopCount ≥ maxHops
timestamp	long Unix ms; required	Creation time at source; NOT updated by relay nodes; used for order
seenBy	String\[\] JSON array	Ordered list of group IDs that processed this message; prevents inter-g
priority	int 1-4; set by SRP	1=SOS, 2=LOCATION, 3=CHAT, 4=HEARTBEAT; governs PriorityBlockingQueue o
batteryLevel	int 0-100; optional	Sender's battery % at transmission time; displayed in topology and

Table 2.6: SRP message structure – all fields, types, and descriptions

```
// Complete SRP SOS message JSON – A3F2-9C1B broadcasts emergency |
| :---- |
| { |
|   "messageId"   : "550e8400-e29b-41d4-a716-446655440000", |
|   "senderId"    : "A3F2-9C1B", |
|   "senderName"  : "ZOUBIRI", |
|   "recipientId" : "BROADCAST", |
|   "messageType" : "SOS", |
|   "payload"     : "35.7034,4.5521|BATTERY:23%|NEED\_HELP", |
|   "hopCount"    : 0, |
|   "maxHops"     : 10, |
|   "timestamp"   : 1712345678901, |
|   "seenBy"      : \["GRP\_A"\], |
|   "priority"    : 1, |
|   "batteryLevel": 23 |
```

```
| }\<\<EOF\>\> |
```

2.6 Store-and-Forward Mechanism

When the SRP routing layer determines that the target device (recipientId) is not currently reachable – no

```
| // SharedPreferences-backed queue persistence (JSON) |
| :---- |
| @Entity(tableName \= "pending\_messages") |
| public class StoreForwardMessage { |
|     @PrimaryKey |
|     public String messageId;          // UUID – prevents duplicate storage |
|     @ColumnInfo(index \= true) |
|     public String recipientId;        // indexed for fast lookup on reconnection |
|     public String serializedJson;     // complete SRPMessage JSON string |
|     public long   storedAt;           // Unix ms – for expiry logic |
|     public int    retryCount;         // max 5 retries before permanent drop |
|     public int    priority;           // preserved from SRPMessage (SOS=1 first) |
| } |

| Column | Type | Purpose |
| ----- | :---: | ----- |
| messageId | TEXT PRIMARY KEY | Unique message ID; prevents duplicate queue entries |
| recipientId | TEXT NOT NULL INDEX | Target device; indexed for O(1) lookup on BLE rediscovery |
| serializedJson | TEXT NOT NULL | Full JSON SRPMessage; deserialized to SRPMessage on delivery |
| storedAt | INTEGER NOT NULL | Unix epoch ms; messages older than 30 min dropped by expiry job |
| retryCount | INTEGER DEFAULT 0 | Incremented per failed delivery attempt; dropped after 5 retries |
| priority | INTEGER NOT NULL | 1=SOS always delivered before CHAT on reconnect; preserves original priority
```

Table 2.9: Store-and-forward database schema (pending_messages table)

On BLE rediscovery of a previously offline device, StoreForwardQueue.flush(recipientId) retrieves all pending

2.7 Formal UML Models

2.7.1 Use Case Diagram

Two human actors interact with SPAN RESCUE: the Rescue Worker (operational coordinator, sends/receives mes

- UC1 – Form Mesh Network (automated): includes FR1 (BLE), FR2 (WFD), TCP initialization; precondition for
- UC2 – Send SOS Alert (Victim/Worker → system): extends UC1; includes GPS attach, P=1 queue, 3× repeat;
- UC3 – Send Chat Message (Worker): extends UC1; includes JSON serialize, P=3 queue, SRP relay;
- UC4 – Relay Message (MeshService): automated; triggered by any message with hopCount \< maxHops; extends
- UC5 – View Live Topology (Worker): reads NodeRegistry from MeshService; extends UC1;
- UC6 – Monitor System Phases (Worker): reads 7-phase lifecycle state from MeshService; extends UC1;
- UC7 – Store Undeliverable Message (MeshService): extends UC4; triggered when recipientId unreachable; in
- UC8 – Deliver Queued Messages (MeshService): triggered by BLE rediscovery; includes flush(recipientId),

2.7.2 Sequence Diagram – BLE Discovery and Group Formation

```
| «Device A launches SPAN RESCUE» |
| :---- |
| MeshService.onCreate() |
|     → BLEDiscovery.startAdvertising(id=A3F2, status=IDLE, bat=84) |
|     → BLEDiscovery.startScanning(onDeviceFound callback) |
|     → restartHandler.postDelayed(restartTask, 28\_000) |
| |
| «Device B launches SPAN RESCUE» |
| MeshService.onCreate() |
|     → BLEDiscovery.startAdvertising(id=9C1B, status=IDLE, bat=71) |
|     → BLEDiscovery.startScanning(onDeviceFound callback) |
| |
| «\~1-3 seconds: B detects A's BLE advertisement» |
```

```

| BLEDiscovery.onDeviceFound(deviceId=A3F2) |
|   → WifiDirectManager.discoverPeers() |
|   → PEERS\_CHANGED\_ACTION received |
|   → WifiDirectManager.connect(peerAddress=A) |
|   → CONNECTION\_CHANGED\_ACTION received on A and B |
|   → WifiDirectManager.discoverPeers() ← CRITICAL re-call |
|   → WifiDirectManager.requestGroupInfo() |
| |
| «GO election: A wins (higher Intent Value)» |
|   A: isGroupOwner=true, IP=192.168.49.1 |
|     → TCPSocketManager.startServer(port=8765) |
|   B: isGroupOwner=false, goIP=192.168.49.1 |
|     → TCPSocketManager.connectToGO(192.168.49.1, 8765\)|
| |
| «TCP connection established» |
|   SRP pipeline active on both devices |
|   Heartbeat exchange begins every 5 seconds |
|   UI updated: MESH\_ACTIVE | 2 nodes | phase 4 complete |

```

2.7.3 Sequence Diagram – SOS Transmission and Multi-hop Relay

```

| «User presses SOS button → 2-second long-press confirmed» |
| :---- |
|   SOSButton.onLongClickConfirmed() |
|     → LocationManager.getLastKnownLocation() → lat=35.7034, lng=4.5521 |
|     → SRPMessage.create(type=SOS, recipient=BROADCAST, |
|       payload='35.7034,4.5521|BATTERY:23%', |
|       hopCount=0, maxHops=10, priority=1) |
|     → seenCache.put(messageId) |
|     → priorityQueue.offer(msg, P=1) ← inserted at HEAD |
|     → UI: SOS pulse animation starts, banner displayed |
| |
| «MeshService forwarding thread – SOS (P=1, delay=0ms)» |
|   tcpManager.broadcast(msg) → all connected clients \[repeat 3x\] |
|   bridgeForwarder.forward(msg) → adjacent groups |
| |
| «Device B (1 hop away) receives SOS» |
|   TCP read → EOF buffer extracts JSON |
|   SRP.process(msg): |
|     seenCache: MISS → add messageId |
|     recipientId=BROADCAST → deliverToUI() |
|     hopCount++ \= 1 \< 10 → priorityQueue.offer(P=1) |
|   UI: full-screen red SOS overlay shown |
|   Audio: emergency alert sound plays |
|   B forwards to its neighbors (hop=2) |
| |
| «Device C (2 hops away) receives SOS» |
|   SRP.process: seenCache MISS → process → deliver → forward (hop=3) |
| |
| «Device D (3 hops away) receives SOS» |
|   SRP.process: seenCache MISS → process → deliver → TTL check: 3\<10 → forward |

```

2.7.4 Class Diagram – Core Networking Components

```

| MeshService (extends Service) \[FR7 – background operation\] |
| :---- |
| ■ bleManager : BLEDiscoveryManager |
| ■ wfdManager : WifiDirectManager |
| ■ tcpManager : TCPSocketManager |
| ■ srpRouter : SmartRoutingProtocol |
| ■ sfQueue : StoreForwardQueue |
| ■ bridgeFwd : BridgeForwarder |
| ■ nodeRegistry : ConcurrentHashMap\<String, NodeInfo\> |
| \+ onStartCommand(intent, flags, id): int |
| \+ onDestroy(): void // releases all BLE, WFD, TCP, DB resources |

```

```

|   ■ buildNotification(): Notification |
|   |
| BLEDiscoveryManager |
|   ■ advertiser   : BluetoothLeAdvertiser |
|   ■ scanner      : BluetoothLeScanner |
|   ■ restartHandler: Handler (Looper.getMainLooper()) |
|   ■ restartTask   : Runnable (28s periodic) |
|   \+ startAdvertising(id, name, status, bat): void |
|   \+ startScanning(cb: DiscoveryCallback): void |
|   \+ stopAll(): void |
|   |
| WifiDirectManager |
|   ■ wfpManager    : WifiP2pManager |
|   ■ channel        : WifiP2pManager.Channel |
|   ■ receiver       : WifiDirectBroadcastReceiver |
|   \+ createGroup(): void |
|   \+ requestGroupInfo(listener): void |
|   \+ connect(deviceAddress: String): void |
|   \+ discoverPeers(): void // called after EVERY connection event |
|   \+ releaseChannel(): void // called in onDestroy() |
|   |
| TCPSocketManager |
|   ■ serverSocket  : ServerSocket      \[GO only – port 8765\] |
|   ■ bridgeSocket  : ServerSocket      \[Bridge GO – port 8889\] |
|   ■ clientSocket  : Socket            \[Clients only\] |
|   ■ executor      : ThreadPoolExecutor(5, 19\)|
|   ■ buffers       : Map\<Socket, StringBuilder\> \[per-client EOF buffer\] |
|   \+ startServer(port: int): void |
|   \+ connectToGO(ip: String, port: int): void |
|   \+ send(socket: Socket, msg: SRPMessage): void |
|   \+ onRawData(socket: Socket, data: String): void |
|   |
| SmartRoutingProtocol |
|   ■ seenCache     : synchronized LinkedHashMap(500, LRU) |
|   ■ pQueue        : PriorityBlockingQueue\<SRPMessage\> |
|   ■ fwdExecutor   : ScheduledExecutorService |
|   \+ process(msg: SRPMessage): void |
|   \+ forward(msg: SRPMessage): void // 0-500ms jitter delay |
|   ■ getPriority(type: MessageType): int |
|   ■ deliverToUI(msg: SRPMessage): void // LocalBroadcastManager |
|   |
| StoreForwardQueue |
|   ■ queueKey      : String             [shared_preferences key] |
|   \+ enqueue(msg: SRPMessage): void |
|   \+ flush(recipientId: String): List\<SRPMessage\> |
|   \+ expireOld(maxAgeMs: long): void // drops messages \> 30 min old |
|   |
| SRPMessage implements Comparable\<SRPMessage\> |
|   \+ messageId, senderId, senderName, recipientId |
|   \+ messageType: MessageType |
|   \+ payload, hopCount, maxHops, timestamp |
|   \+ seenBy: List\<String\> |
|   \+ priority, batteryLevel |
|   \+ compareTo(other): int // by priority ASC (lower \= higher priority) |

```

2.8 User Interface Design

2.8.1 Design System Tokens

```

| Token Name | Hex Value | Application |
| ----- | :---: | ----- |
| color-background | \#020810 | All screen backgrounds – near-black |
| color-surface | \#0D1B2A | Cards, panels, bottom navigation bar |
| color-accent-cyan | \#00F5FF | Active states, icons, borders, mesh topology edges |
| color-emergency-red | \#FF2244 | SOS button, SOS messages, critical alert banners |

```

color-safe-green	\#00FF88	Normal node status, connected indicators, success states
color-warning-orange	\#FF8C00	Alert node status, battery <20% warning
color-text-primary	\#FFFFFF	All primary body text and labels
color-text-secondary	\#8899AA	Timestamps, hop counts, secondary info
font-tactical	Courier New / Monospace	All data displays: Device IDs, IPs, hop counts
font-ui	System default sans-serif	Navigation, buttons, general labels
touch-target-min	48dp × 48dp	All interactive elements – WCAG AA, gloved hands
sos-button-diameter	120dp	Dominant focal point; impossible to miss

Table 2.11: SPAN RESCUE UI design system tokens

*** 2.8.2 Screen 1 – Home / SOS**

The home screen centers on the SOS button: a 120dp circular element with a steady red pulse animation in s

\[Home Screen – SOS Button, Mesh Stats, Radar View \]	*← Insert screenshot here →*
:---:	

*Fig. 2.21: UI Wireframe – Home Screen with SOS button (2s long-press), live mesh statistics row, and rada

*** 2.8.3 Screen 2 – System Monitor**

A vertical stepper displaying seven connection phases. Each phase card: phase name, animated status indica

\[System Monitor Screen – 7-Phase Lifecycle Display \]	*← Insert screenshot here →*
:---:	

Fig. 2.22: UI Wireframe – System Monitor showing seven phases with phase 4 (TCP) active

*** 2.8.4 Screen 3 – Mesh Topology**

A Canvas-rendered live graph. Nodes: circles (28dp radius) with colored borders: cyan=G0, green=Normal cli

\[Mesh Topology Screen – Live Node Graph \]	*← Insert screenshot here →*
:---:	

Fig. 2.23: UI Wireframe – Mesh Topology with 5 nodes: ME (G0/cyan), 3 clients (green), 1 bridge (teal)

*** 2.8.5 Screen 4 – Tactical Chat**

Standard bubble layout (sent=right/cyan, received=left/dark). Each bubble: sender name+ID, message text, t

\[Tactical Chat Screen – Hop-Count Display and SOS Overlay \]	*← Insert screenshot here →*
:---:	

Fig. 2.24: UI Wireframe – Tactical Chat with HOP:2 badge and full-width SOS banner at top

*** 2.8.6 Screen 5 – Settings**

Device name edit (max 16 chars), Device ID display (read-only XXXX-YYYY), maxHops slider (3–15, default 10)

2.9 Conclusion**

This chapter has presented the complete design and formal conception of SPAN RESCUE across six dimensions:

Chapter 3: Implementation and Testing

3.1 Introduction

This chapter describes the realization of the architecture from Chapter 2 as a working Android application

> Note: The code snippets in this chapter are conceptual pseudo-code illustrating the architecture. The ac

3.2 Development Environment

Component	Specification and Rationale
----	-----
IDE	Android Studio Hedgehog 2023.1.1 – latest stable at development time
Programming Language	Dart (Flutter) for application logic, with Kotlin for native Android foreground s
Target SDK	Android 14 (API 34\)
Minimum SDK	Android 10 (API 29\)
Build System	Gradle 8.0 \+ Android Gradle Plugin 8.1
JSON Serialization	Dart `dart:convert` JSON encoding/decoding; newline-delimited JSON over TCP sockets

```
| Local Database | `shared_preferences` persistent JSON queue; single-key storage for pending messages |
| UI Framework | Flutter widgets + Material Design 3; native Android code limited to foreground service and
| Background Execution | Foreground Service (MeshService) with FOREGROUND\_SERVICE\_TYPE\_CONNECTED\_DEVICE
| Service → UI IPC | Flutter state management / MethodChannel; UI updates performed on the Flutter main isolate
| Threading Model | Dart async/await with periodic `Timer` tasks and stream-based event handling; native Kotlin
| Version Control | Git – private GitHub repository; feature branches per component |
| Test Devices | 5 Android smartphones: 4 manufacturers, API levels 29/31/33/34 |
```

Table 3.1: Development environment and build configuration

3.3 MeshService – Foreground Service Architecture

MeshService is the single most architecturally critical decision in SPAN RESCUE. Android's background execution

```
| // AndroidManifest.xml – Service declaration |
| :---- |
| \<service |
|     android:name="com.spanrescue.tactical.MeshForegroundService" |
|     android:exported="false" |
|     android:foregroundServiceType="connectedDevice|location" /\> |
| |
| // Conceptual Android Foreground Service initialization – actual implementation is split between Dart and Kotlin
| @Override |
| public void onCreate() { |
|     super.onCreate(); |
|     // RULE: startForeground() MUST be first call (Android 14 requirement) |
|     startForeground(NOTIFICATION\_ID, buildPersistentNotification()); |
| |
|     deviceId = loadOrGenerateDeviceId(); // XXXX-YYYY, stored in SharedPreferences |
|     // In the real app, BLE discovery, Wi-Fi Direct negotiation, and TCP transport are handled in Flutter
|     // The native service primarily maintains the persistent notification and wake lock. |
| }
|
| sfQueue    \= new StoreForwardQueue(this); |
| bridgeFwd  \= new BridgeForwarder(tcpManager); |
|
| bleManager.startAdvertising(deviceId, userName, STATUS\_IDLE, getBattery()); |
| bleManager.startScanning(this::onDeviceDiscovered); |
| wfdManager.registerReceiver(); |
| } |
|
| @Override |
| public void onDestroy() { |
|     bleManager.stopAll(); |
|     wfdManager.releaseChannel(); // MUST release to prevent Wi-Fi Direct leak |
|     tcpManager.closeAllSockets(); |
|     sfQueue.expireOld(30 \* 60 \* 1000L); |
|     super.onDestroy(); |
| } |
```

3.4 Android Permissions

```
| Permission | Min API | Purpose and Notes |
| :---- | :---: | :---- |
| ACCESS\_FINE\_LOCATION | 29 | Mandatory for Wi-Fi Direct peer discovery since Android 10 – not documented |
| CHANGE\_NETWORK\_STATE | All | Enable/disable Wi-Fi network state for WFD management |
| ACCESS\_WIFI\_STATE | All | Read WFD group information and peer list |
| CHANGE\_WIFI\_STATE | All | Initiate WFD group formation, connection, and disconnection |
| BLUETOOTH\_ADVERTISE | 31 | BLE advertising – new split permission model introduced in Android 12 |
| BLUETOOTH\_SCAN | 31 | BLE scanning – split permission; also requires neverForLocation=true for non-local |
| BLUETOOTH\_CONNECT | 31 | BLE device management – required for getRemoteDevice() calls |
| FOREGROUND\_SERVICE | All | Required to call startForeground(); prevents service termination |
| FOREGROUND\_SERVICE\_CONNECTED\_DEVICE | 34 | Foreground service using BLE/WFD – mandatory since Android 14 |
| WAKE\_LOCK | All | PARTIAL\_WAKE\_LOCK held only during TCP TX burst; released immediately after – prevents
```

Table 3.2: Android permissions required – purpose and minimum API level

3.5 Critical Implementation Findings – Undocumented Android Behaviors

The following four behaviors were discovered exclusively through real-device testing across five smartphones

****Finding 1: Wi-Fi Direct Peer Discovery Auto-Stop After Every Connection****

Android's Wi-Fi Direct subsystem automatically stops peer discovery after every connection attempt – both

```
| // WifiDirectBroadcastReceiver.java – INCORRECT (common mistake) |
| :---- |
| case WifiP2pManager.WIFI_P2P_CONNECTION_CHANGED_ACTION: |
|     wfdManager.requestGroupInfo(callback); |
|     // ← Missing discoverPeers() here – device STOPS discovering permanently |
|     break; |
| |
| // CORRECT implementation – must re-call after every connection event |
| case WifiP2pManager.WIFI_P2P_CONNECTION_CHANGED_ACTION: |
|     wfdManager.requestGroupInfo(callback); |
|     wfdManager.discoverPeers(); // ← CRITICAL: restart discovery |
|     break; |
```

****Finding 2: BLE Scan Auto-Termination After ~30 Seconds in Background****

Android's power management terminates BLE scan operations after approximately 30 seconds when the application

```
| // BLEDiscoveryManager.java – 28-second periodic scan restart |
| :---- |
| private final Handler restartHandler = new Handler(Looper.getMainLooper()); |
| private final Runnable restartTask = new Runnable() { |
|     @Override public void run() { |
|         if (!isScanning) return; |
|         bleScanner.stopScan(scanCallback); |
|         SystemClock.sleep(200); // brief pause to reset scanner state |
|         bleScanner.startScan(filters, settings, scanCallback); |
|         Log.d(TAG, "BLE scan restarted (28s keepalive)"); |
|         restartHandler.postDelayed(this, 28_000); // reschedule before 30s kill |
|     } |
| }; |
| // Called from startScanning(): |
| restartHandler.postDelayed(restartTask, 28_000); |
```

****Finding 3: GO Election Non-Determinism Across Device Manufacturers****

Wi-Fi Direct GO election is non-deterministic across different manufacturers even when Intent Values are e

```
| // WifiDirectManager.java – role-adaptive initialization (CORRECT) |
| :---- |
| wfdManager.requestGroupInfo(channel, group) { |
|     if (group == null) { |
|         Log.w(TAG, "Group info null – retrying in 2s"); |
|         retryHandler.postDelayed(() -> wfdManager.requestGroupInfo(...), 2000); |
|         return; |
|     } |
|     if (group.isGroupOwner()) { |
|         Log.i(TAG, "Role: GROUP OWNER | IP: 192.168.49.1"); |
|         tcpManager.startServer(PORT_PRIMARY); // ServerSocket(8765) |
|         tcpManager.startBridgeServer(PORT_BRIDGE); // ServerSocket(8889) |
|     } else { |
|         String goIP = "192.168.49.1"; // always fixed for GO |
|         Log.i(TAG, "Role: CLIENT | GO: " + goIP); |
|         tcpManager.connectToGO(goIP, PORT_PRIMARY); |
|     } |
| }); |
```

****Finding 4: TCP Fragmentation at Messages Above ~200 Bytes****

On all 5 test devices, TCP messages of 200 bytes or more were regularly fragmented across multiple read()

```
| // INCORRECT – naive single-read parsing (fails ~15-40% of messages >200B) |
```



```
| :---- |
| final data = await socket.first; |
| final json = utf8.decode(data, allowMalformed: true); // may be partial JSON – CORRUPT |
| final msg = SRPMessage.fromJson(jsonDecode(json) as Map<String, dynamic>); |
| |
| // CORRECT – line-delimited stream parsing |
| socket.map((data) => utf8.decode(data, allowMalformed: true)) |
|   .transform(const LineSplitter()) |
|   .listen((line) { |
|     try { |
|       final msg = SRPMessage.fromJson(jsonDecode(line) as Map<String, dynamic>); |
|       srpRouter.process(msg); |
|     } catch (e) { |
|       // malformed message dropped, but parsing remains resilient |
|     } |
|   }); |
```

3.6 Application Screenshots

The following screenshots were captured during a live testing session on a Samsung Galaxy A52 (Android 13,

```
| \[ Fig. 3.1 – Home Screen (insert your screenshot here) \] *← Insert screenshot here →* |
| :---: |
```

*Fig. 3.1: SPAN RESCUE Home Screen – SOS button in standby (pulsing red ring), 5 nodes connected, mesh sta

```
| \[ Fig. 3.2 – System Monitor (insert your screenshot here) \] *← Insert screenshot here →* |
| :---: |
```

*Fig. 3.2: System Monitor Screen – Phases 1-4 green (complete); Phase 5 SRP cyan+animated (active, 47 msg

```
| \[ Fig. 3.3 – Mesh Topology (insert your screenshot here) \] *← Insert screenshot here →* |
| :---: |
```

*Fig. 3.3: Mesh Topology Screen – ME node (cyan/GO) at center; 3 client nodes (green, HOP:1); 1 bridge no

```
| \[ Fig. 3.4 – Tactical Chat (insert your screenshot here) \] *← Insert screenshot here →* |
| :---: |
```

*Fig. 3.4: Tactical Chat Screen – sent messages (cyan right), received messages (dark left) with HOP:2|VIA

```
| \[ Fig. 3.5 – Settings Screen (insert your screenshot here) \] *← Insert screenshot here →* |
| :---: |
```

*Fig. 3.5: Settings Screen – Device Name: ZOUBIRI; Device ID: A3F2-9C1B (read-only); maxHops: 10 (slider 3

3.7 Test Results

3.7.1 Test Device Specifications

ID	Device Model	Android Ver.	Wi-Fi Chip	Role in Tests
D1	Samsung Galaxy A52	Android 13 (API 33\)	Qualcomm QCA6390	Group Owner – most sessions
D2	Xiaomi Redmi Note 10	Android 12 (API 31\)	MediaTek MT7668	Client / Bridge node
D3	Huawei Y9 Prime	Android 10 (API 29\)	HiSilicon Hi1103	Client – minimum API validation
D4	OnePlus 9	Android 13 (API 33\)	Qualcomm QCA6391	Client / SOS sender
D5	Samsung Galaxy A32	Android 12 (API 31\)	Qualcomm QCA6390	Store-and-forward target

Table 3.3: Test device specifications – 5 devices, 4 manufacturers, 3 Android versions

3.7.2 Test Session Results

Test Scenario	Result	Key Metric	Observation
2-device direct link (1 hop, 100 msgs)	✓ PASS	Avg. 145ms	SRP \+ TCP overhead negligible; 0 duplica
5-device chain relay (4 hops, 100 msgs)	✓ PASS	All delivered	SeenCache: 0 duplicates across all 10
SOS priority under load (50 CHAT queued)	✓ PASS	SOS always first	SOS delivered before any CHAT mes
SOS repeated 3× reliability guarantee	✓ PASS	99.7% delivery	3× repeat overcame 1-packet loss; near
GO failure \+ automatic self-healing	✓ PASS	Reform in avg. 16.3s	BLE rediscovery → new GO electio
Store-and-forward delivery (5 min offline)	✓ PASS	100% delivery	Messages queued correctly; SOS de

TTL enforcement (maxHops=10)	✓ PASS	0 infinite loops	All messages dropped exactly at hop=10; no br
BLE background scan (2h continuous)	✓ PASS	Scan active full 2h	28s restart maintained discovery; C
TCP fragmentation (200-500B messages)	✓ PASS	0 corrupted msgs	EOF buffer assembled all fragments;
Cross-manufacturer compatibility	✓ PASS	5/5 devices	All 4 manufacturers interoperated; Finding 3 v
Network formation time (10 sessions)	✓ PASS	Avg. 4.1s	Min: 2.8s; Max: 6.9s; all within 10s requir
SeenCache duplicate prevention (rapid fire)	✓ PASS	0 duplicates	100 rapid messages: 0 duplicates

Table 3.4: Test session results – all scenarios and outcomes

3.7.3 Post-Implementation Comparative Analysis

Capability	SPAN RESCUE	FireChat	Bridgefy	Meshtastic	SGN \[7\]
SOS priority \+ 3x repeat	✓ P=1,3x	✗	✗	Partial	Partial
Multi-group bridge \>8 devices	✓ 20 dev.	✗	✗	✓	✗
Store-and-Forward (shared_preferences JSON)	✓	✗	✗	✗	✗
Live topology \+ hop display	✓	✗	✗	✓	✗
Network self-healing \<20s	✓	✗	✗	✓	✗
Background via ForegroundSvc	✓	Partial	Partial	✓	Partial
TTL broadcast storm protection	✓ (3 mech)	Unknown	Unknown	✓	✗
Standard hardware only	✓	✓	✓	✗ (LoRa)	✓
Open and auditable	✓	✗	✗	✓	Research
Undocumented API documented	✓ (4 findings)	✗	✗	N/A	✗
End-to-end security	Planned	Unknown	Vuln. \[11\]	✓	Partial

Table 3.5: Post-implementation comparison with related SPAN systems

3.8 Conclusion

The implementation of SPAN RESCUE confirms that the architecture designed in Chapter 2 is fully realizable

General Conclusion

This thesis set out to answer a concrete, life-critical question: can a group of standard Android smartphones communicate in a secure, reliable, and scalable manner? The three chapters each contribute a distinct and complementary layer to this answer. Chapter 1 established the problem and the goals of the research. Chapter 2 presented the architecture and implementation of SPAN RESCUE. Chapter 3 presented the results of the implementation and the comparative analysis. Beyond the application itself, the four undocumented Android API behaviors documented in Section 3.5 constitute a significant contribution to the understanding of Android's internal workings.

Summary of Contributions

- A four-layer communication architecture (BLE discovery \+ Wi-Fi Direct star groups \+ TCP with EOF framing)
- The Smart Routing Protocol (SRP): controlled flooding with LRU deduplication (500 entries), TTL enforcement
- A multi-group software bridge topology scaling to 20 devices across 4 star groups with automated inter-group
- A complete Android application with five tactical screens, Foreground Service architecture, shared_preferences
- Documentation of four undocumented Android Wi-Fi Direct and BLE API behaviors with tested, production-ready
- Validation across five devices from four manufacturers confirming cross-platform compatibility and all t

Limitations

- Security: message encryption is not yet implemented. The current SRP transmits JSON in plaintext, vulner
- Scale validation: full 20-device physical deployment was not possible due to hardware availability; the
- GPS dependency: SOS payloads include GPS coordinates; indoor or underground scenarios where GPS is unava

Future Work

- Authenticated key exchange: implement ECDH session key negotiation for end-to-end AES-256-GCM encryption
- Adaptive TTL: dynamically estimate mesh diameter from observed hop counts and adjust maxHops accordingly
- Health sensor integration: extend the SOS payload to include biometric data from BLE wearables (heart ra
- Large-scale simulation: validate the 20-device topology and SRP performance beyond physical hardware ava
- iOS interoperability: extend to iOS using the Multipeer Connectivity Framework, enabling mixed Android/i
- Field deployment: conduct a structured field test with a university search-and-rescue simulation team to

SPAN RESCUE demonstrates, in working code tested on real devices, that the wireless hardware already prese

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